

FROM ONTOGENY TO INSTITUTIONAL CALCIFICATION

Group Membership Bias as the Cognitive Engine of Asymmetric Legal Persistence

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Abstract

Why do legal institutions persist even when evidence overwhelmingly favors reform? The standard explanations, from veto players (Tsebelis, 2002) to path dependence (Pierson, 2000), describe outcomes without explaining their cognitive mechanics.

Recent experimental evidence offers a candidate mechanism: Confer et al. (2025) demonstrate that children as young as four alter their evidentiary standards as a function of group membership, sampling less evidence before adopting group beliefs ($\chi^2(1) = 78.15$, $p = 0.012$), inflating confidence in group-supporting evidence by approximately 39% ($\chi^2(1) = 128.03$, $p < 0.001$), and revising beliefs at half the rate when group identity is salient ($\chi^2(1) = 7.46$, $p = 0.006$).

If this cognitive architecture, which operates before metacognitive control develops, persists into adulthood and scales through professional socialization, it could supply the missing microfoundation for institutional persistence.

I propose Heteronomous Bayesian Updating (HBU) as a formalization of this scaling mechanism and organize its institutional effects through a multilevel game-theoretic typology.

The framework generates falsable predictions that I test illustratively against the Constitutional Lock-in Index (CLI) developed in prior work. Argentina's Article 14bis, with a CLI of 0.87 and 0/23 successful reform attempts between 1991 and 2025, serves as the primary case.

The core claim is theoretical: legal institutions calcify not because of conspiracy or path dependence alone, but because the human cognitive apparatus for evaluating evidence is systematically distorted by group affiliation, and legal professions constitute precisely the kind of identity-conferring groups that trigger this distortion.

Keywords: group membership bias, institutional persistence, extended phenotype theory, Bayesian updating, constitutional lock-in, legal reform, cognitive development, evidentiary standards

JEL Classification: K00, K31, D83, D72, Z13

1. THE PUZZLE OF DISPROPORTIONATE PERSISTENCE

Some legal institutions resist change with a ferocity that existing theories struggle to explain. Brazil reformed its *Consolidação das Leis do Trabalho* in 2017 after decades of failed attempts. Chile overhauled labor regulation six times between 1990 and 2023. Spain restructured its constitutional labor framework through five major reforms between 1994 and 2022. Argentina, during the same period, attempted 23 labor reforms; not one produced sustained structural change (Lerer, 2024a).

The conventional toolkit offers three frameworks for understanding this asymmetry. Tsebelis (2002) counts veto players; more players mean more policy stability. Pierson (2000) identifies increasing returns; early institutional choices create self-reinforcing dynamics. Olson (1965) diagnoses collective action failures; concentrated beneficiaries defeat diffuse losers. Each captures something real. None explains why the same constitutional architecture produces lock-in in Argentina but not in structurally similar jurisdictions with comparable numbers of veto players, similar colonial legal heritage, and analogous interest group configurations.

The missing variable, I argue, is cognitive. Specifically, the mechanism documented by Confer et al. (2025) in a study published in *Nature Communications*: group membership systematically alters how humans evaluate evidence, and it does so from a developmental stage before conscious strategic reasoning becomes available.

The theoretical contribution proceeds in three steps. First, I document the experimental findings and their implications for institutional theory (Section 2). Second, I formalize a scaling mechanism, Heteronomous Bayesian Updating, that connects individual cognitive bias to collective institutional behavior (Section 3). Third, I organize the institutional manifestations through a multilevel game-theoretic typology and generate falsable predictions (Section 4). Section 5 offers an illustrative application through the Constitutional Lock-in Index.

2. THE EXPERIMENTAL FOUNDATION

2.1 What Confer et al. Actually Show

Joshua Confer and colleagues at UC Berkeley conducted three preregistered studies with 262 children aged 4 to 6. The experimental design was elegant in its simplicity: children either joined one of two minimal groups or learned about two groups without joining either. All children then faced the same evidentiary reasoning task, determining whether a set of ten boxes contained more elephants or lions. The boxes were arranged so that the first five seemed to support one conclusion, but the full set of ten supported the opposite.

The results were striking across all three studies.

Study 1 (N = 78): Truncated evidence search. Children who belonged to a group opened significantly fewer boxes before reaching a conclusion ($\chi^2(1) = 78.15$, $p = 0.012$, $R^2 = 0.08$). Group-affiliated children opened approximately four boxes on average; unaffiliated children opened approximately six. The consequence was dramatic: roughly a quarter of group-condition children reached the correct conclusion, compared with more than half of no-group children ($\chi^2(1) = 8.53$, $p = 0.003$, $R^2 = 0.14$). Children who belonged to a group were, in

the authors' words, "half as likely to hold an accurate belief about the boxes." A mediation analysis revealed that 47.1% of the effect on beliefs operated through premature termination of evidence search (ACME = -0.119, 95% CI [-0.238, -0.011], $p = 0.030$). The direct effect of group membership on belief, once the mediator was controlled, was not significant (ADE = -0.133, $p = 0.186$).

Study 2 (N = 124): Inflated confidence. When shown identical evidence (two boxes), children in the group condition reported substantially higher confidence that the evidence supported their group's belief: $M = 7.16$ ($SD = 2.62$) versus $M = 5.13$ ($SD = 3.16$) on a nine-point scale ($\chi^2(1) = 128.03$, $p < 0.001$, $R^2 = 0.11$). The inflation amounts to approximately 39%, calculated as $(7.16 - 5.13) / 5.13$. Same evidence, same task, same age; the only difference was whether the child had been assigned to a group moments earlier.

Study 3 (N = 60): Resistance to counterevidence. After forming an initial belief, children were shown two pieces of opposing evidence. No-group children reduced their confidence by an average of 4.2 units ($SD = 3.9$); group children reduced it by only 2.2 units ($SD = 2.6$) ($\chi^2(1) = 60.00$, $p = 0.019$, $R^2 = 0.09$). The effect on binary belief revision was even starker: 68% of no-group children switched their initial belief after counterevidence, compared with 34% of group children ($\chi^2(1) = 7.46$, $p = 0.006$, $R^2 = 0.18$). Group membership halved the rate at which children updated beliefs in response to disconfirming data.

Two features of these results deserve emphasis. First, the age range matters: children at four years old, before they develop metacognitive strategies for evaluating their own reasoning, already exhibit biased evidence processing under group membership. The bias is not learned through socialization into specific group norms; it is triggered by mere minimal group assignment. Second, the authors found no significant interaction between age and condition across any of the three studies. The effect did not increase with age in this sample, although the range (4 to 6) may have been too narrow to detect developmental change; an older-is-stronger pattern was reported in related work by Roberts et al. (2017).

2.2 The Scaling Hypothesis

Confer et al. demonstrated the mechanism in preregistered laboratory experiments with preschoolers. The extrapolation I propose, that the same cognitive architecture operates in adult legal professionals embedded in professional identity groups, is a hypothesis, not a demonstrated fact. It is the central theoretical claim of this paper.

The hypothesis rests on three bridging premises, each of which requires independent empirical validation:

Premise 1: Persistence. If group membership bias operates before metacognitive control develops (as shown in 4-year-olds), it plausibly represents a foundational feature of social cognition rather than a developmental phase that education eliminates. Adult studies on partisan motivated reasoning (Kahan, 2013; Kahan et al., 2012) and identity-protective cognition (Van Bavel & Pereira, 2018) are consistent with persistence but do not prove developmental continuity.

Premise 2: Amplification through professional identity. Legal education constitutes a prolonged, high-intensity group formation process. Law students acquire not just knowledge but professional identity, normative commitments, and affiliation with doctrinal schools (Mertz, 2007). If minimal group assignment produces measurable bias in four-year-

olds within minutes, years of professional socialization should, at minimum, not reduce the effect.

Premise 3: Institutional embedding. Professional biases aggregate through institutional structures: specialized courts, bar associations, law faculties, professional journals. These structures are not merely neutral conduits; they are, in the language of Dawkins (1982), extended phenotypes of the doctrinal memes they propagate and defend. Each structure selects for professionals who share the group's epistemic commitments, creating self-reinforcing communities of biased evaluators.

If all three premises hold, the laboratory effect documented in children would not merely persist in adult professionals but would be amplified and institutionalized, producing the disproportionate resistance to reform that the Constitutional Lock-in Index measures.

3. FORMALIZING THE MECHANISM: HETERONOMOUS BAYESIAN UPDATING

3.1 From Rational Bayesian Agents to Social Bayesian Actors

Standard institutional economics models agents as Bayesian updaters who incorporate new evidence according to its likelihood given prior beliefs:

$$\frac{P(H|E) = P(E|H) \cdot P(H)}{P(E)}$$

Confer et al.'s data suggest this formalization misses something. Children in the no-group condition behaved approximately as Bayesian theory predicts: they sampled evidence adequately and updated beliefs in proportion to evidential weight. Children in the group condition did not. They truncated evidence search, overweighted group-consistent evidence, and underweighted disconfirming evidence.

I propose that group membership introduces a weighting function $w(A)$ that distorts evidence evaluation as a function of group affiliation A :

$$P_a(H|E) = \frac{P(E|H)^{w(A)} \cdot P(H)}{P(E)}$$

Where:

- $w(A) > 1$ when evidence supports the group belief (overweighting, as in Study 2)
- $w(A) < 1$ when evidence opposes the group belief (discounting, as in Study 3)
- $w(A) = 1$ when no group membership is salient (rational baseline, as in all no-group conditions)

I call this **Heteronomous Bayesian Updating (HBU)**, borrowing Kant's distinction between heteronomous action (governed by external authority) and autonomous action (governed by self-legislated reason). The term captures the core mechanism: **belief revision becomes responsive not just to evidence but to social affiliation, and it does so through automatic cognitive processes rather than conscious strategic choice.**

3.2 Properties and Limitations of HBU

The formalization is deliberately minimal. It does not specify the functional form of $w(A)$, its distributional properties, or how it interacts with prior strength. These are empirical questions that require investigation beyond what Confer et al.'s design can answer. What the formalization does accomplish is making the theoretical claim precise enough to be falsable:

Prediction H1: Professional groups with stronger identity markers (specialized vocabulary, credentialing requirements, formal membership) should exhibit higher $w(A)$ values, measurable through experimental variants of the Confer paradigm adapted for adult professionals.

Prediction H2: The magnitude of bias should correlate with institutional measures of group cohesion, such as bar association density, specialized court concentration, and professional journal self-citation rates.

Prediction H3: Reforms that bypass professional evaluation, such as legislative overrides enacted during political windows, should succeed more often than reforms requiring professional consensus, holding substantive content constant.

HBU is also not without theoretical alternatives. Hierarchical Bayesian models (Gelman et al., 2013) can capture group-level priors without introducing a separate weighting function. Identity-based utility models (Akerlof & Kranton, 2000) can explain group-conforming beliefs through preference modification rather than cognitive distortion. Distinguishing HBU from these alternatives requires experiments designed specifically to tease apart cognitive and motivational mechanisms. Confer et al.'s data are consistent with, but do not uniquely support, the cognitive interpretation.

3.3 Relationship to Extended Phenotype Theory

In previous work (Lerer, 2024b, 2025a), I argued that legal institutions function as extended phenotypes of cultural replicators, memes in Dawkins' (1982) technical sense. Legal doctrines build material structures (courts, agencies, professional associations) that ensure doctrinal transmission and defense, just as beaver genes build dams that modify the environment to favor beaver gene propagation.

HBU supplies a candidate cognitive mechanism for this process. Dawkins demonstrated that extended phenotypes require no conscious intention; the beaver does not intend to create a dam that benefits its genes. Similarly, **HBU suggests that institutional lock-in requires no conspiracy. Legal professionals do not consciously decide to overweight evidence supporting their doctrinal commitments. The bias is automatic, pre-metacognitive in origin, and amplified by professional socialization. The result is coordinated institutional defense without explicit coordination, precisely the pattern that extended phenotype theory predicts.**

4. A MULTILEVEL GAME-THEORETIC TYPOLOGY

4.1 Type 1 Games: Constitutional Interpretation

Constitutional provisions are interpreted by professionals who have spent years forming group identities around specific doctrinal traditions. In the HBU framework, judges and constitutional scholars do not merely apply their training; their training has calibrated their $w(A)$ function so that group-consistent interpretations of ambiguous text receive evidential overweighting.

Consider Argentina's Article 14bis, which guarantees workers' rights through deliberately vague language (e.g., "participación en las ganancias de las empresas, con control de la producción y colaboración en la dirección"). Each interpretive community, laboristas, constitucionalistas, privatistas, processes the same constitutional text through different $w(A)$ calibrations. The HBU framework predicts that interpretive competition will produce asymmetric outcomes: expansive interpretations backed by established professional communities will prove more resistant to revision than restrictive interpretations lacking comparable institutional support.

This prediction is consistent with the observed pattern. Argentina's Supreme Court has systematically expanded Article 14bis protections since the 1957 constitutional reform, with the doctrine of *ultraactividad* (indefinite survival of expired collective bargaining agreements) emerging as a canonical example. Each expansion creates new professional infrastructure (specialized labor courts, labor law chairs, professional journals) that amplifies the $w(A)$ of the expansive interpretation, making further expansion easier and reversal harder.

Falsable prediction T1: In jurisdictions where constitutional labor provisions are interpreted by generalist courts rather than specialized labor courts, the rate of doctrinal expansion should be lower and the success rate of reformist reinterpretation should be higher, controlling for constitutional text similarity.

4.2 Type 2 Games: Regulatory Compliance and Enforcement

Regulatory enforcement involves a second layer of HBU: enforcement professionals form group identities around specific regulatory philosophies ("tough on crime," "protective of workers," "pro-competition") that distort how they evaluate evidence of compliance or violation.

The mechanism generates testable predictions for regulatory outcomes. If enforcement professionals systematically overweight evidence consistent with their regulatory identity and discount evidence favoring deregulation, regulated entities face an asymmetric enforcement environment: compliance failures are detected and punished at rates exceeding what neutral evidence evaluation would produce, while evidence of regulatory burden or overenforcement is discounted. This could explain why, across multiple jurisdictions, regulatory intensity tends to increase monotonically over time. Regulatory sunset provisions fail not because legislators forget to review, but because the professional communities charged with review evaluate evidence of regulatory necessity through group-biased lenses.

Falsable prediction T2: Enforcement agencies staffed by professionals with longer tenure and stronger professional identity markers should exhibit higher false positive rates (detecting violations where none exist) and lower false negative rates (missing genuine violations), compared with agencies staffed by generalists or rotating appointees.

4.3 Type 3 Games: Professional Socialization and Doctrinal Reproduction

The deepest level of institutional lock-in operates through education and professional socialization, where the next generation of legal professionals acquires the $w(A)$ calibrations that will govern their evidence evaluation for decades.

Law schools do not merely transmit information; they form professional identities. The learning method, clinical programs, moot courts, and faculty mentorship all operate as group formation mechanisms analogous to (though vastly more intense than) the minimal group paradigm in Confer et al.'s experiments. If minimal group assignment biases four-year-olds' evidence evaluation within minutes, years of immersive professional socialization should produce correspondingly stronger $w(A)$ calibrations.

The implication for institutional evolution is significant: **legal education functions as the reproductive mechanism of doctrinal memes, calibrating each generation's cognitive apparatus for evidence evaluation in group-consistent ways.** This is not a metaphor; it is a specific empirical claim about how professional training modifies evidence processing, a claim that the Confer paradigm could test directly if adapted for adult professional populations.

Falsable prediction T3: Law students' evidence evaluation, measured through Confer-style paradigms adapted for legal reasoning, should show increasing group bias across years of legal education, with specialization (e.g., labor law track) producing larger effects than general legal training.

5. ILLUSTRATIVE APPLICATION: THE CONSTITUTIONAL LOCK-IN INDEX

5.1 From Description to Explanation

In prior work (Lerer, 2024a, 2024b, 2024c), I developed the Constitutional Lock-in Index (CLI) to quantify institutional resistance through four empirical indicators: precedent density (P), measured by citation networks and interpretive entrenchment; doctrinal elaboration (D), measured by law review volume and scholarly consensus; professional organization (O), measured by bar association structure and certification requirements; and enforcement infrastructure (E), measured by specialized tribunals and administrative capacity:

$$\text{CLI} = 0.30 \times P + 0.25 \times D + 0.20 \times O + 0.25 \times E$$

Applied to Argentina's Article 14bis, the CLI yielded 0.87 on a zero-to-one scale. For comparison: Brazil's CLT scored 0.40 before its successful 2017 reform, Chile's Labor Code scored 0.24, and Spain's constitutional labor framework scored 0.37. Across 23 Latin American reform attempts between 1990 and 2024, CLI scores predicted reform outcomes with 78% accuracy; reforms targeting provisions with $\text{CLI} > 0.70$ failed in 19 of 20 attempts (95% failure rate), while reforms targeting provisions with $\text{CLI} < 0.45$ succeeded in 14 of 18 attempts (78% success rate).

Those results were descriptive and predictive. The CLI measured lock-in without explaining *why* high scores predict resistance. HBU supplies the missing mechanistic account. Each CLI component measures a different dimension of the institutional infrastructure through which group-biased evidence evaluation transmits across professional generations:

- **Precedent density (P)** captures how deeply the $w(A)$ calibration has been embedded in judicial decision-making. Higher precedent density means more authoritative "evidence" weighted through group-biased lenses.
- **Doctrinal elaboration (D)** measures the extent to which scholarly production has created a self-referential evidential ecosystem. Scholars citing scholars who share their doctrinal commitments produce an evidence base that appears overwhelming when evaluated through group-biased cognition.
- **Professional organization (O)** quantifies the strength of group identity markers. Bar associations, certification requirements, and professional networks are the adult equivalents of Confer et al.'s minimal group assignment, except they operate through years of immersive socialization rather than minutes of arbitrary assignment.
- **Enforcement infrastructure (E)** measures the institutional capacity to enact group-biased evidence evaluation in practice. Specialized courts staffed by career judges trained in the same doctrinal tradition provide the material substrate for $w(A)$ -distorted adjudication.

5.2 The Argentine Case as Illustration

Argentina's labor regime represents an extreme case: CLI = 0.87, with 187 specialized labor courts, mandatory labor law certification for practitioners, 23 failed reform attempts across seven administrations and three decades. The HBU framework predicts this outcome without requiring conspiracy theories about union power or cultural explanations about Argentine attitudes toward labor protection.

The mechanism is simpler and more parsimonious. Argentina's constitutional design (vague text, treaty incorporation at constitutional rank, specialized judicial infrastructure) created conditions maximally favorable for HBU amplification. Each successful expansion of labor protections generated new professional infrastructure that selected for professionals sharing the expansive interpretation. Each generation of labor lawyers and judges was socialized into an increasingly cohesive professional identity, calibrating their evidence evaluation toward overweighting evidence of worker vulnerability and underweighting evidence of regulatory burden.

The result is an institutional system that resists reform not because its defenders are irrational or corrupt, but because they process evidence through cognitive architecture that has been shaped, from early childhood, to privilege group-consistent information. The ultraactividad doctrine, under which expired collective bargaining agreements continue in force indefinitely, is not an irrational anomaly; it is the predictable phenotypic expression of a doctrinal meme defended by a professional community whose evidence evaluation has been systematically calibrated to favor its persistence.

5.3 Cross-National Variation as Natural Experiment

If HBU is the operative mechanism, cross-national variation in reform success should correlate with variation in professional group cohesion rather than with variation in political institutions or economic conditions alone. Brazil's successful 2017 labor reform (Lei 13.467/2017) was enacted during a period of political crisis with unusually low professional consensus, precisely the conditions under which HBU predicts reduced institutional

resistance. Chile's multiple successful reforms coincided with a legal profession historically less specialized and less corporatized than Argentina's, consistent with lower $w(A)$ calibrations.

These observations are suggestive, not conclusive. Rigorous cross-national testing requires controlling for political variables (coalition size, presidential power, electoral timing), economic conditions (crisis severity, unemployment rates, fiscal pressure), and institutional design features (amendment requirements, judicial review scope) before attributing residual variation to cognitive mechanisms. I am developing a dataset of reform attempts across multiple jurisdictions to enable such analysis, but the data collection is ongoing and the results are not yet available.

6. THEORETICAL POSITIONING

6.1 What HBU Adds to Existing Frameworks

The claim is not that Tsebelis, Pierson, and Olson are wrong. Veto players, path dependence, and collective action dynamics all operate in institutional evolution. The claim is that they are incomplete without a microfoundational account of how individual decision-makers process evidence within institutional settings.

Tsebelis counts veto players but assumes each player evaluates proposals through stable, exogenously given preferences. HBU suggests preferences are partly endogenous to group membership, which means the number of veto players understates the difficulty of reform when professional groups are cohesive and overstates it when they are fragmented.

Pierson identifies increasing returns but does not explain the cognitive mechanism generating them. HBU supplies this mechanism: each expansion of institutional infrastructure recalibrates the $w(A)$ function of the professionals who staff it, making further expansion cognitively easier and reversal cognitively harder. Increasing returns are not merely structural; they are cognitive.

Olson explains why concentrated beneficiaries defeat diffuse losers but does not explain why beneficiaries so consistently overestimate the value of existing arrangements and underestimate the evidence for reform. HBU offers an answer: concentrated beneficiaries, precisely because they form cohesive professional groups, process reform evidence through systematically biased cognitive filters.

6.2 Limitations and Unresolved Questions

Honesty requires acknowledging what this framework does not accomplish.

First, the developmental continuity assumption is untested. Confer et al. studied 4- to 6-year-olds. No published study, to my knowledge, has replicated their specific paradigm with adult professionals. The adult literature on motivated reasoning is consistent with the HBU hypothesis but does not test it directly.

Second, the formalization of $w(A)$ is schematic. The weighting function has no estimated parameters, no specified functional form, and no direct measurement protocol. Until these gaps are filled, HBU functions as a theoretical organizing principle rather than a quantitative model.

Third, the CLI validation, while suggestive (78% predictive accuracy across 23 cases), relies on coding decisions that could be contested. The weighting of components (0.30/0.25/0.20/0.25) was derived from regression analysis on a limited sample and has not been cross-validated on independent data.

Fourth, alternative mechanisms could produce similar institutional patterns. Professional self-interest (rather than cognitive bias) could explain resistance to reform. Cultural transmission of normative commitments (rather than calibration of evidence evaluation) could produce doctrinal persistence. Distinguishing HBU from these alternatives requires experimental designs that Confer et al.'s work makes possible but that no one has yet conducted.

7. A RESEARCH AGENDA

The theoretical framework generates a research program with three tiers:

Tier 1: Developmental continuity. Replicate the Confer paradigm with law students at different stages of legal education and with practicing professionals in different specializations. The prediction is specific: group bias in evidence evaluation should be measurable in law students and should correlate with year of study and degree of specialization.

Tier 2: Cross-national institutional analysis. Compile a comprehensive dataset of legal reform attempts across jurisdictions with varying levels of professional group cohesion. The prediction is that CLI scores, interpreted through the HBU lens, will outperform purely structural or political predictors of reform success.

Tier 3: Intervention design. If HBU is the operative mechanism, interventions reducing the salience of professional group identity during policy evaluation should improve reform prospects. This could include blind review of reform proposals (removing information about which professional community supports or opposes them), rotating judicial assignments to prevent specialization-induced group formation, and structured deliberation protocols that separate evidence evaluation from group affiliation.

Each tier generates predictions that are falsable with existing methodological tools. The framework's value lies not in its certainty but in its capacity to organize empirical inquiry around a specific, testable mechanism.

IA USE DISCLAIMER – ACKNOWLEDGMENT

Statistical analysis for the CLI validation was conducted with computational assistance from Claude (Anthropic) and verified through independent calculation. All theoretical frameworks, legal interpretations, and empirical claims are the sole responsibility of the author. AI assistance was limited to data processing and literature review support.

The Confer et al. (2025) statistics reported in Section 2 were verified directly against the published article and its supplementary materials.

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Replication Materials

Complete replication materials for the Constitutional Lock-in Index, including datasets, coding procedures, and statistical analysis scripts, are available at: <https://github.com/adrianlerer/Argentina-Labor-Regime-Analysis-2025>

Concept Paper for Zenodo. Version 2.0. Computational assistance: Claude (Anthropic). All theoretical claims and legal interpretations are the sole responsibility of the author.